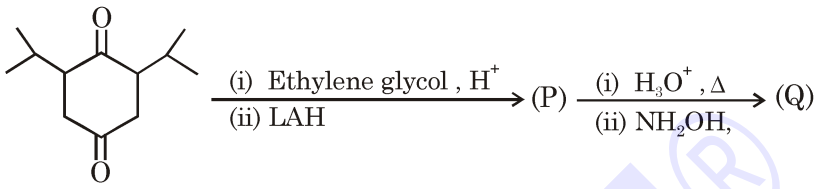
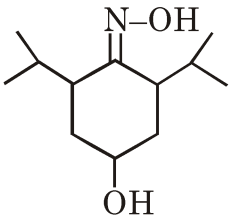
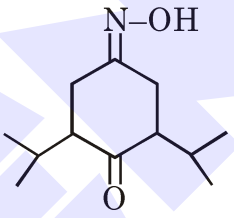
**SECTION–I : (i) Only One option correct Type**

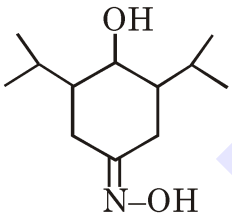
This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **3(–1)**

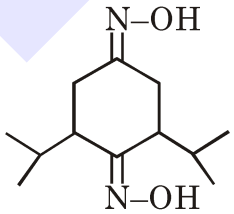
**1.** Major product (Q) in following sequence is :

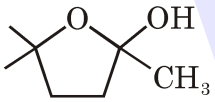



(A) 

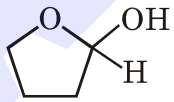

(B) 


(C) 


(D) 


**2.** 

Compound-I



Compound-II

Correct statement about compound-I & compound-II is :

(A) Compound-I gives Tollen's test with Tollen's reagent & Compound-II gives Iodoform test with NaOH, $I_2$

(B) Compound-I gives Iodoform test with NaOH, $I_2$ & Compound-II gives Tollen's test with Tollen's reagent

(C) Both comopunds gives Tollen's test with Tollen's reagent

(D) Both comopunds gives Iodoform test with NaOH, $I_2$



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**3.** Starch $\xrightarrow{HOH / H^+}$ mixture (A)

Glycogen $\xrightarrow{HOH / H^+}$ mixture (B)

Cellulose $\xrightarrow{HOH / H^+}$ mixture (C)

Which of the following mixture contain β-D-glucose, after treating following polymers with $H_3O^{\oplus}$ for long time.

(A) Only in mixture A (B) Only in mixture B

(C) Only in mixture C (D) In all mixtures A, B & C

**4.** Two acyclic molecules having molecular formula $C_5H_8$ one of which shows enantiomerism & the other shows diastereoisomerim. The two acyclic molecule are related as :

(A) Chain Isomers (B) Positional Isomers

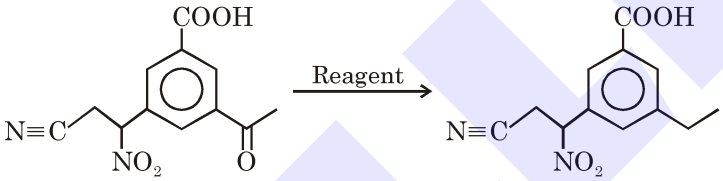
(C) Geometrical Isomers (D) Conformation isomers

**5.** A certain carboxylic acid X which can be isolated form whale blubber or sardine oil on ozonolysis with $O_3$ / Zn / $H_2O$ yields nonanal & 5-oxo pentanoic acid. The correct option regarding X is :

(A) It is example of β-γ unsaturated acid (B) Its molecular formula is $C_{14}H_{28}O_2$

(C) This acid does show geometrical isomerism (D) This acid exists as (±) enantiomeric pair

**6.**




Above conversion is carried out with the help of :

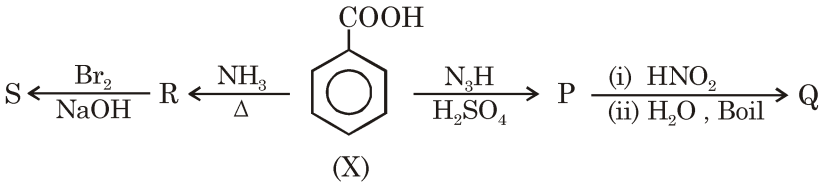
(A) $LiAlH_4$ (B) $NaBH_4$

(C) $NH_2$–$NH_2$ / $OH^{\ominus}$ / Δ , then $H^{\oplus}$ (D) Zn–Hg / HCl

## SECTION-I(ii) : One or More than one

This section contains **6 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE** are correct. **4(–1)**

**7.** For the following sequence, choose the correct option(s) ?




(A) (Q) will give positive $Br_2$, $H_2O$ test

(B) (R) on reaction with $NaNO_2$, HCl produces (X)

(C) Major product (P) & (S) are identical

(D) Reaction sequence involves schmidt & Hoffmann bromamide degradation reaction



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**8.** Which of the following amino acid(s) has isoelectric point more than 7

(A) Aspartic acid (B) Phenyl alanine

(C) Arginine (D) Histidine

**9.** Which set of reagents can carry out following conversion ?

$$HC{\equiv}CH \longrightarrow CH_3{-}CH_2{-}CH_2{-}CH_3$$

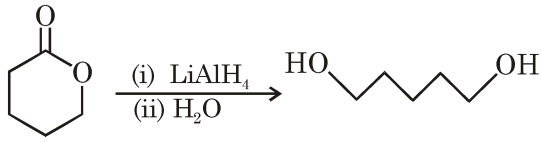
(A) (i) $HgSO_4$, $H_2SO_4$ ; (ii) NaOH, $\Delta$ ; (iii) $N_2H_4$, $H_2O_2$ ; (iv) Zn–Hg / HCl

(B) (i) $NaNH_2$ (2 mole) ; (ii) $CH_3$–I (2 mole) ; (iii) $H_2$ / Pd , $BaSO_4$ ; (iv) $N_2H_4$, $H_2O_2$

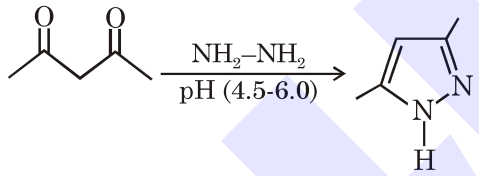
(C) (i) $H_2$ / Pd , $BaSO_4$ ; (ii) HBr / $H_2O_2$ ; (iii) Na / $Et_2O$

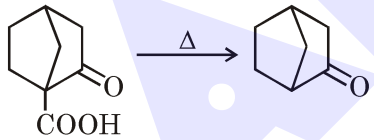
(D) (i) $B_2H_6$ ; (ii) $H_2O_2$, $\overset{\ominus}{O}H$ ; (iii) Mg-Hg / $H_2O$ ; (iv) Red P / HI

**10.** Which of the following reaction(s) are correctly matched with their major product ?

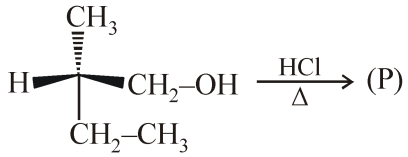
(A) 


(B) $CH_3{-}CHO + 3Br_2 + 4NaOH \longrightarrow HCOONa + CHBr_3 + 3NaBr + 3H_2O$

(C) 


(D) 


**11.** For the following reaction, choose the incorrect option(s) ?




(–)-2-methylbutane-1-ol

(A) Formation of major product (P) involves carbocation rearrangement

(B) Major product (P) is 2-methyl but-2-ene

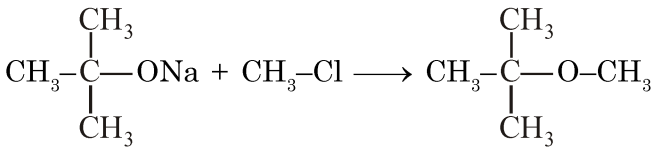
(C) Reaction follows bimolecular nucleophilic substitution

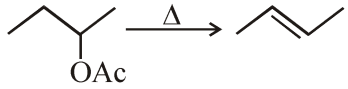
(D) Major product (P) is 2-chloro-2-methyl butane

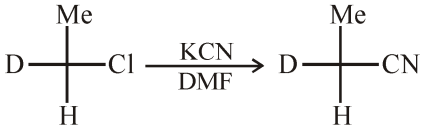


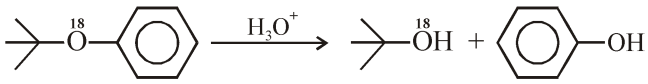
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**12.** Which of following reactions are shown with incorrect major product :

(A) 

(B) 

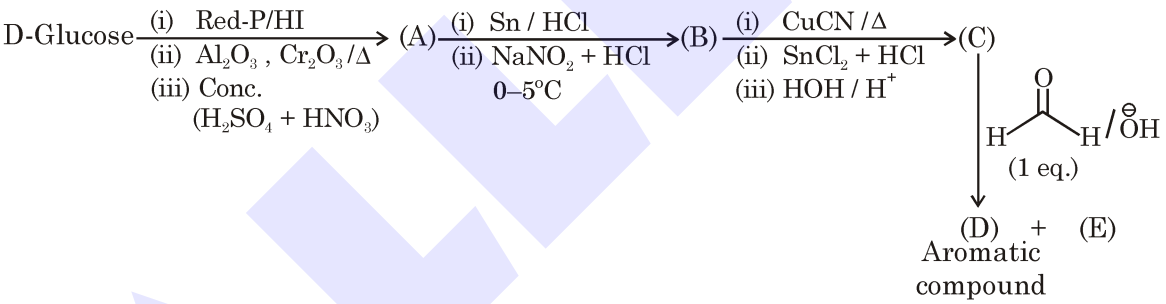
(C) 

(D) 

## SECTION–I(iii) : Paragraph Type

This section contains **1 paragraphs** each describing theory, experiment, data etc. **3 questions** relate to two paragraphs with two questions on each paragraph. Each question of a paragraph has **only one correct answer** among the four choices (A), (B), (C) and (D). **3(0)**

### Paragraph for Q.No. 13 to 14




**13.** Select correct statement among following :

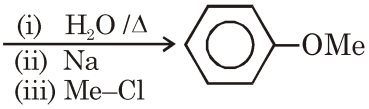
(A) Compound 'C' on treating with conc. NaOH show disproportionation reaction

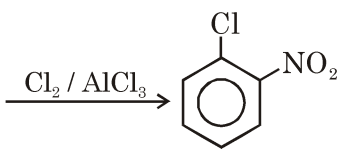
(B) Compound 'C' form diazonium salt with aq. $HNO_2$ or ($NaNO_2$ + HCl) / 0–5°C

(C) Compound 'C' give foul smelling compound with ($CHCl_3$ + KOH)

(D) Compound 'C' is primary amine

**14.** Select the correct statement / reaction among following :

(A) Comopund-B 

(B) Compound-A 

(C) Compound-D is $Ph–CH_2–N=CH_2$

(D) Compound-E is $Ph–C\overset{\ominus}{O}O\overset{\oplus}{N}a$

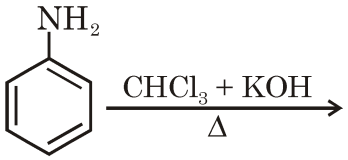


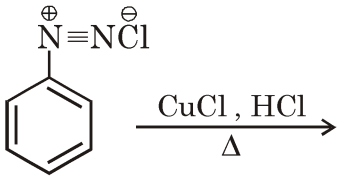
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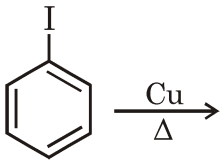
## SECTION–I(iv) Matching List Type

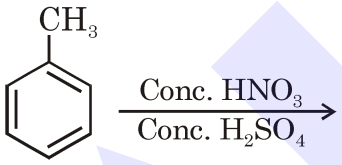
This Section contains **1 multiple choice questions. Each question has matching lists.** The codes for the lists. have choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct. **4(0)**

**15.** **List-I** **List-II**

(P)  (1) Aryl halide is obtained as a product

(Q)  (2) It is electrophilic substitution reaction

(R)  (3) Reaction is isocyanide test

(S)  (4) New carbon-carbon bond is formed in product

**Code :**

| | P | Q | R | S |
|---|---|---|---|---|
| (A) | 4 | 1 | 2 | 3 |
| (B) | 3 | 1 | 4 | 2 |
| (C) | 1 | 2 | 4 | 3 |
| (D) | 3 | 1 | 2 | 4 |

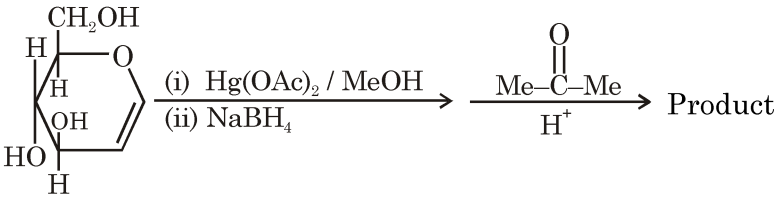

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## SECTION–III : Integer Type

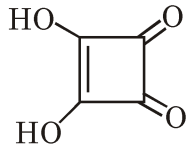
This section contains **4 questions**. The answer to each question is **a single digit Integer**, ranging from **0 to 9** (both inclusive) **4(–1)**

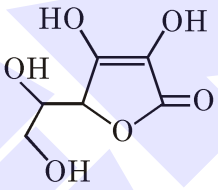
**1.** "X" dichloro derivatives are possible for n-butane. What will be the value of (X – 3) :

**2.** Identify total number of chiral centre present in product of following reaction :

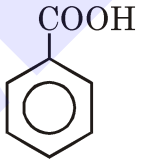



**3.** Identify total number of compounds which librate $CO_2\uparrow$ gas on reaction with aqueous $NaHCO_3$?

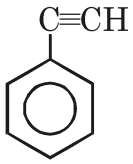
(a) 


(b) 


(c) 


(d) 


(e) 


(f) 


(g) $H_2SO_4$

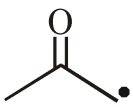
(h) MeCOOH

**4.** Identify total number of free radicals which have $sp^2$ hybridised central atom (having free radical) :

















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## SECTION–IV : Matrix Match

This section contains **ONE** questions. Each question contains two columns, **Column-I** and **Column-II. Column-I** has **four** entries (A), (B), (C) and (D) **Column-II** has **five** entries (P), (Q), (R), (S) and (T) Match the entries in **Column-I** with the entries in **column-II.** One or more entries in **Column-I** may match with one or more entries in **Column-II.** The ORS contains a 4 × 5 matrix whose layout will be similar to the one shown below :

(A) (P) (Q) (R) (S) (T)
(B) (P) (Q) (R) (S) (T)
(C) (P) (Q) (R) (S) (T)
(D) (P) (Q) (R) (S) (T)

---

**1.**

| **Column - I** | **Column - II (One of the organic product can)** |
|---|---|
| (A) But-2-ene $\xrightarrow[\text{(iv) } CrO_3, H_2SO_4]{\text{(i) } O_3, \text{(ii) } Zn, H_2O \text{ (iii) dil. NaOH}}$ | (P) Evolve $CO_2$ on heating |
| (B) $CH_3–CH=O$ + HO, HO $\xrightleftharpoons{H^+}$ | (Q) Cyclic gemdiether |
| (C) $CH_3CHO \xrightarrow[\text{(ii) } H_3O^+]{\text{(i) KCN/HCl}}$ | (R) Give cyclic ester product on heating |
| (D)  $\xrightarrow[\text{(ii) } H_2O_2]{\text{(i) } O_3}$ | (S) Evolve $CO_2$ with aq. $NaHCO_3$ |
| | (T) Can give diol on reduction with $LiAlH_4$ |


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